Essay 1: current paper in one slide

Question and setting

Do firm website restrictions on search and discovery crawlers change market liquidity around mandatory disclosures?

- Treatment: the first restriction of any of 31 search/discovery crawlers in archived monthly `robots.txt` files.
- Not an AI paper: the treatment covers the full search/discovery layer, including conventional search bots.
- Design: staggered adoption from 2021 to 2025; 27,327 10-K/10-Q filing events for 1,908 firms.
- Main outcome: filing-window log Amihud, measured over $[-1, +1]$, relative to the firm's local benchmark $[-60, -10]$.

Current headline result

BJS estimate: **+0.0134**, $p = 0.010$. The TWFE estimate is **+0.0109**.

- The filing window does not become less liquid in levels; its local benchmark improves faster.
- Quoted spreads are null, dollar volume rises, and return variance rises.
- Effects are concentrated among smaller, low-institutional-ownership, and low-coverage firms.

Current interpretation

The evidence points toward price impact/depth or trader composition, not wider quoted spreads or trading withdrawal.

Feedback from the committee and our discussion

1. Measurement paper

Candidate contribution: the first systematic evidence that firms **actively make their own public information harder for machines to discover**.

This requires a first stage in actual discoverability, not only a stated `robots.txt` policy.

2. Mechanism

Amihud is not information deterioration.

Separate lower information acquisition, more informed trading, noise-trader exit, and changes in disclosure content.

3. More output variables

Expand the outcome set, but require each variable to adjudicate among mechanisms.

Priority families: price response, information timing, investor breadth, monitoring, and disclosure behavior.

The dissertation studies effective, not merely legal, publicity

Common theoretical primitive

$$\text{Effective public information} = \text{legal availability} \times \text{discoverability} \times \text{processing capacity}$$

Information can be legally public and still behave like private information if outsiders cannot find it cheaply and repeatedly. The three essays study a different equilibrium margin of this same friction:

Trading margin

Who trades, how much depth they supply, and how strongly prices move per unit of volume.

Monitoring margin

Who continuously reads firm information and how managers respond when monitoring becomes costly.

Intermediation margin

Whether open-web discovery substitutes for or complements traditional human intermediaries.

Conceptual anchors: Grossman and Stiglitz (1980), Merton (1987), Kyle (1985), and Dugast and Foucault (2018).

Three essays share a friction, not an outcome swap

	Essay 1	Essay 2	Essay 3
Working title	Crawler-Access Restrictions and Disclosure-Window Liquidity	Turning Off the Machines	Searchability and Information Intermediary Substitution
Literature home	Market microstructure	Corporate governance	Information intermediaries
Economic margin	Trading and price impact	External monitoring	Substitution vs. complementarity
Main shock	Firm website crawler restriction	Firm website crawler restriction	Broker closure or research-department shutdown
Role of discoverability	Treatment and measured friction	Treatment plus monitoring exposure	Predetermined moderator, not treatment
Primary object	Liquidity and the timing of price learning	Opportunistic managerial behavior	Damage caused by exogenous analyst loss

Important correction

Essay 2 still uses the robots restriction shock. It is independent only if it establishes a distinct monitoring channel and a prespecified governance outcome. Essay 3 genuinely changes the treatment.

Essay 1 measures a firm choice over discoverability

What the measure captures

- A monthly policy governing whether named crawlers may request content from the firm's website.
- A firm's ability to change the machine accessibility of information that remains visible to a human visitor.
- A broad search/discovery infrastructure choice, not a narrow response to generative AI.

What it does not yet prove

- That major crawlers obeyed the restriction at the relevant time.
- That indexed pages, ranking, or discovery latency actually changed.
- That managers personally intended the change rather than inheriting a vendor or security template.

Careful contribution language

Until intent and the actual discovery first stage are validated, the defensible statement is: **firms' websites impose restrictions that reduce permitted machine access**. "Firms deliberately hide" is the stronger hypothesis to be tested.

Essay 1 uses filing events to isolate a high-information window

Empirical design

- Unit: firm filing event for 10-K and 10-Q disclosures.
- Treatment date: first observed restriction of any focal search/discovery crawler.
- Comparison: matched never-restricting firms and not-yet-treated observations.
- Estimators: BJS imputation as the main design; conventional TWFE as a transparent comparison.
- Outcome: abnormal filing-window log Amihud relative to a firm-local pre-filing benchmark.

Why use a local benchmark?

The estimand compares the filing window with the same firm's recent normal trading environment. This removes much of the persistent cross-sectional difference in baseline liquidity.

Interpretation constraint

A positive abnormal measure may arise because the event window worsens, because normal days improve, or both. The two components must be shown separately.

The main effect is relative deterioration, not a level collapse

Main estimate

+0.0134

BJS abnormal log Amihud

$p = 0.010$

Concentration

The effect is stronger for small firms, firms with low institutional ownership, and firms with little analyst coverage.

What moves underneath the ratio?

- Filing-window Amihud does not rise materially in levels.
- The non-event benchmark improves faster after restrictions.
- Dollar volume rises rather than falls.
- Quoted spreads do not widen.
- Return variance rises around filings.
- Post-filing drift and delayed-response share are currently null.

Most accurate current reading

The filing window becomes relatively more price-impact intensive. The evidence does not yet distinguish lower depth, concentrated information arrival, and trader composition.

Amihud can increase for economically different reasons

The measure

$$\text{Amihud}_{it} = 10^6 \frac{|r_{it}|}{\text{DollarVolume}_{it}}$$

Channel	Why Amihud rises	Additional implication
Lower market depth	A given order imbalance moves price more	Effective-spread or price-impact proxies rise; breadth may fall
More informative trades	Returns respond more strongly to fundamentals	Return-news covariance rises, even if liquidity is unchanged
Noise traders exit	Denominator falls because uninformed volume disappears	Volume and participation breadth fall
Information becomes concentrated	More of the quarterly news arrives at the filing	Pre-filing learning falls; filing response rises
Local benchmark changes	Normal-day liquidity improves faster than event liquidity	Abnormal measure rises without an event-window level increase

Therefore

The paper cannot infer “information worsens” from Amihud. It must identify the component and price-learning pattern that changed.

Essay 1 needs a mechanism horse race, not a larger outcome menu

Mechanism	Volume / breadth	Spread / depth	Fundamental response	Post-filing learning
Information loss	May fall	May worsen	Full-cycle response falls	Drift or delayed response rises
Temporal concentration	Need not fall	Event price impact may rise	Pre-filing down; filing up; total unchanged	Little residual learning
Informed-trader composition	Volume may rise or stay flat	Price impact rises more than quoted spread	Response per trade rises	No necessary drift
Noise-trader exit	Volume and breadth fall	Price impact may rise	Fundamental mapping need not change	No sharp prediction
Disclosure-content change	Depends on news quantity and complexity	Depends on content	Changes are explained by filing news	Depends on omitted content

How the current facts narrow the race

Higher dollar volume and null quoted spreads make simple trading withdrawal less plausible. Higher return variance with no post-filing drift is consistent with either concentrated information arrival or a change in price impact. The next tests must distinguish these two.

Theoretical motivation: Kyle (1985); Merton (1987); Davila and Parlatore (2026) on the non-monotone relation between trading costs and price informativeness.

“Where Prices Learn” belongs inside Essay 1

Its role in the paper

It explains **why the same mandatory filing may generate greater price movement per unit of trading after search access is restricted**.

Before restriction

Search and discovery tools expose continuous website signals: IR updates, product news, hiring, operational announcements, and changes to firm pages.
Some quarterly information may therefore enter prices before the filing date.

After restriction

The cost of finding continuous signals rises. The mandatory SEC filing remains accessible and may become a larger information reset.
Price learning may shift from dispersed pre-filing days to the filing window.

This is a mechanism for the Essay 1 market result. With the same restriction events, firms, filing cycle, and return data, it would look like salami slicing if presented as a separate essay.

The relevant question is when and through which channel prices learn

PRE-FILING
Continuous web signals
Anticipatory incorporation

$\Longrightarrow$

FILING
Mandatory disclosure
Scheduled information reset

$\Downarrow$

POST-FILING
Residual learning
Delay or correction?

Terminology

"Where" does not mean a physical trading venue. It means **where in the information cycle** and **through which disclosure channel** fundamental news enters prices. A more precise section title is: *When and Through Which Channel Do Prices Learn?*

Price learning requires a fundamental anchor

Primary measures

- Window-specific earnings response coefficients using standardized unexpected earnings.
- Covariance between current returns and future earnings innovations.
- Horizon-specific price informativeness in the spirit of Bai, Philippon, and Savov.
- Full-cycle price accuracy relative to subsequently realized fundamentals.

Diagnostic measures only

- Absolute returns and return variance.
- Hou-Moskowitz price delay.
- Variance ratios and return synchronicity.
- Post-earnings-announcement drift.

Measurement hierarchy

Lead with return-fundamental mapping. Use variance, delay, and drift to interpret the timing and persistence of learning, not as substitutes for price informativeness.

Methodological anchors: Hou and Moskowitz (2005); Bai, Philippon, and Savov (2016); Davila and Parlatore (2026).

A window-by-window design separates reallocation from destruction

For $w \in \{\text{pre-filing, filing, post-filing, full cycle}\}$, estimate:

Core specification

$$R_{iq}^{w} = \alpha_w + \beta_w \mathit{News}_{iq} + \theta_w\left(\mathit{Restricted}_{iq} \times \mathit{News}_{iq}\right) + \Gamma_w X_{iq} + \mu_i + \tau_q + \varepsilon_{iq}^{w}$$

Scenario	Pre-filing	Filing / post-filing	Full-cycle implication
Temporal reallocation	$\theta_{pre} < 0$	$\theta_{filing} > 0$, little drift	$\sum_w \theta_w \approx 0$
Information destruction	$\theta_{pre} < 0$	Filing response does not compensate; later learning persists	$\sum_w \theta_w < 0$
Depth / composition only	Little change in return-news coefficients	Amihud or price impact rises	Total fundamental mapping is unchanged

Interpretive payoff

If filing-day response rises for the same news, part of the Amihud result is information concentration in the numerator, not simply worse liquidity.

Essay 1 revision should proceed through three gates

Gate 1: discovery first stage

Test whether restriction changes indexed-page counts, search ranking, cache recency, discovery latency, or verified crawler consumption.

EDGAR downloads are an indirect spillover, not the clean first stage.

Gate 2: Amihud decomposition

Show event-window and benchmark components separately; add price-impact/depth proxies, breadth, volume, and spread evidence.

Avoid claiming a level liquidity collapse that the current data do not show.

Gate 3: price learning

Link returns to fundamental news across the pre-filing, filing, and post-filing windows.

Determine whether information is delayed, destroyed, or simply incorporated with greater price impact.

Revised Essay 1 contribution

The paper would establish a corporate choice over discoverability, document its market consequence, and separate depth, composition, and the relocation of price learning toward mandatory disclosure dates.

Essay 2: machine discovery may function as external monitoring

Research question

When a firm restricts machine access to continuously updated public information, does outside monitoring weaken enough to change managerial behavior?

Economic mechanism

Automated discovery is an input into scalable anomaly detection. Restrictions raise monitoring costs, reduce the probability that irregular behavior is noticed, and may increase managers' private benefit from opportunism.

Why this is not Essay 1

The outcome is managerial conduct rather than market liquidity. The mechanism operates through detection and discipline, not through the composition of traders at a disclosure event.

Conceptual precedents: Zhu (2019) on data as governance; Heese, Perez-Cavazos, and Peter (2022) on the loss of local monitoring; Diamond and Verrecchia (1991).

The monitoring channel must focus on information outside mandatory filings

Information plausibly affected

- Investor-relations page updates.
- Product and operating announcements.
- Hiring, location, and organizational changes.
- Corrections, disappearing pages, and other website changes detectable only through repeated crawling.

Information that remains available

- 10-K, 10-Q, 8-K, and Form 4 filings on EDGAR.
- Historical data already collected by institutions.
- Bloomberg, direct feeds, and other proprietary channels.
- Human access to pages that robots are prohibited from crawling.

Required exposure measure

The paper needs a pre-treatment measure of **website-native monitoring dependence**: how much economically relevant information is updated on the corporate site and not promptly mirrored in EDGAR or major feeds.

Opportunistic insider trading should be the primary outcome family

Outcome family	Role in the design	Expected timing
Opportunistic Form 4 trades	**Primary**: frequency, trade direction before news, and post-trade abnormal profits	Can move quickly after monitoring weakens
Routine Form 4 trades	**Within-dataset placebo**: mechanically scheduled trades should respond much less	No systematic change
Accrual and real earnings management	External validity: behavior that requires managerial discretion and accumulates over time	Slower, quarterly or annual response
Reporting discipline	Mechanism extension: NT filings, filing delays, amendments, and restatements	Delayed and potentially asymmetric

Avoid outcome mining

Prespecify opportunistic insider trading as the main family. Use accounting choices and reporting discipline only to test generalization and predicted timing.

Classification template: Cohen, Malloy, and Pomorski (2012).

Essay 2 faces a sharper endogeneity problem than Essay 1

Threat

A firm may restrict crawlers because it anticipates bad news, litigation, website attacks, or opportunistic behavior. Pre-trend tests alone cannot eliminate event-time anticipation.

Design response

- Separate active firm-specific changes from vendor/template waves.
- Control for website redesigns, security incidents, and disclosure changes.
- Compare routine and opportunistic trades within the same filing system.

Identification target

Reuse the BJS staggered framework only after establishing:

1. an actual decline in machine discovery;
2. no pre-trend in insider trading or reporting behavior;
3. stronger effects where website-native monitoring should matter;
4. comparable findings in plausibly passive adoption waves.

Interpretation discipline

High institutional ownership is heterogeneity, not a perfect placebo: proprietary and web-based monitoring may be complements.

Essay 2 opens only if three kill-gates pass

1. Consumption gate

Restrictions must reduce an independently measured form of machine discovery or repeated website monitoring.

2. Pre-trend gate

Opportunistic trading, bad-news indicators, and reporting behavior must not begin changing before restriction.

3. Passive-wave gate

Results must survive separating vendor/template-driven changes from firm-specific active changes.

Why the idea remains attractive

Form 4 and EDGAR reporting outcomes cover small and low-coverage firms that disappear from IBES, 13F, or TAQ-based designs. This aligns the outcome data with the firms for which Essay 1 finds the largest effects.

Main risk

Insider-profit outcomes are noisy and the treatment's causal first stage is currently incomplete. The Form 4 prototype and first-stage pilot should precede full data construction.

Essay 3 asks whether searchability replaces a lost analyst

Research question

When a broker closure or research-department shutdown removes analyst coverage, does strong pre-existing web discoverability buffer the capital-market damage?

Why this design is distinct

The treatment is an external shock to the human intermediary. Discoverability is measured before the event and enters as a moderator. The paper does not estimate another consequence of the firm's robots decision.

Why it solves the coverage problem

Treated firms have analyst coverage by construction. The outcome panel therefore exists exactly where the coverage-loss treatment occurs, avoiding the empty-cell problem in a robots-to-analyst-production design.

Shock-design precedents: Kelly and Ljungqvist (2012); Derrien and Kecskes (2013). Infrastructure precedents: Chang, Ljungqvist, and Tseng (2023); Guo and Mota (2021).

Substitution and complementarity make opposite predictions

Let Q be a market-quality outcome for which higher values are better. Estimate the effect of analyst loss as a function of pre-event discoverability D_i:

Core interaction

$$Q_{it} = \alpha_i + \tau_t + \beta Loss_{it} + \gamma\left(Loss_{it} \times D_i^{pre}\right) + \Gamma X_{it} + \varepsilon_{it}$$

Mechanism	Economic logic	Prediction for γ
Substitution	After the processed analyst channel disappears, investors fall back on searchable raw information. High discoverability reduces the damage.	$\gamma > 0$ for market quality; $\gamma < 0$ if the outcome is illiquidity
Complementarity	Analysts use discoverable raw information as an input. Firms with high discoverability had more productive analyst coverage and lose more when it disappears.	$\gamma < 0$ for market quality; $\gamma > 0$ if the outcome is illiquidity

Raw vs. processed information: Dugast and Foucault (2018).

Predetermined discoverability is useful, but not automatically exogenous

Preferred moderator

A pre-event discoverability index combining:

- crawler permissions;
- indexed-page breadth;
- ranking or official-site visibility;
- cache/update latency;
- share of material information available on the open web.

What “predetermined” does not solve

Discoverability may correlate with firm size, investor-relations quality, disclosure policy, and the strength of the remaining analyst ecosystem. Measuring it before closure prevents reverse timing but does not make it random.

Identification discipline

Compare firms within the same broker-closure event; include broker-event-by-time structure, baseline coverage controls, balance tests, and moderator bins. Call the result treatment-effect heterogeneity unless discoverability is itself plausibly exogenous.

Essay 3 should prioritize information replacement, not Amihud again

Outcome layer	Measures	Purpose
Price informativeness	Future-earnings response, price delay, PEAD, horizon-specific informativeness	Primary evidence that open-web information replaces or complements analysts
Intermediary response	Entry by other analysts, changes in remaining forecast accuracy or dispersion	Shows whether the analyst ecosystem repairs the loss
Information demand	EDGAR or website consumption where valid; media attention	Shows whether investors substitute toward raw sources
Firm response	8-K Item 2.02, earnings releases, and guidance if available	Tests whether firms compensate by supplying more disclosure
Investor base and liquidity	Holding breadth, Amihud, and spread proxies	Secondary capital-market consequences

Boundary from Essay 1

Essay 1 studies a firm restriction. Essay 3 asks whether pre-existing discovery absorbs an exogenous loss of human intermediation. Amihud can be secondary, but should not define the contribution.

Firm disclosure response after coverage loss: Balakrishnan, Billings, Kelly, and Ljungqvist (2014).

The broker-shutdown design has one immediate kill-gate: event count

Two-week feasibility audit

1. Build a 2016-2024 list of broker closures, mergers, and research-department shutdowns.
2. Verify events using public news and map affected analysts in IBES Detail.
3. Count firms, coverage lost, and the low-discoverability interaction cell.
4. Check whether post-event outcomes remain observable for at least four quarters.

Kill criterion

A large total event count is insufficient. The design needs enough firms that both lose meaningful coverage and differ substantially in pre-event discoverability.

Fallback sequence

Expand carefully to broker-level research layoffs or analyst-team departures. If the clean shock remains too thin, transplant the same moderator design to local-newspaper closures rather than forcing a low-powered broker sample.

The three-paper architecture is now cleaner than the earlier version

	Essay 1: Trading	Essay 2: Monitoring	Essay 3: Intermediation
Central question	What happens when firms restrict machine discovery?	Does reduced discovery weaken external discipline?	Can discovery capacity replace a lost analyst?
Treatment	Robots restriction	Robots restriction	Broker-level coverage loss
Primary outcome	Filing-window liquidity and fundamental response timing	Opportunistic insider trading	Price informativeness after analyst loss
Distinct mechanism	Depth, composition, and timing of learning	Detection probability and managerial private benefits	Raw vs. processed information substitution
Main empirical gate	Actual discoverability first stage	Consumption first stage plus no behavioral pre-trend	Sufficient broker-event-by-discoverability cells
Literature community	Microstructure	Governance	Information intermediaries

Thesis-level contribution

Corporate information policy is not exhausted by legal publicity. Its effect also depends on machine discoverability and investors' processing capacity.

Recommended sequence minimizes wasted data work

1. **Run the discoverability pilot first.** This is the life-or-death gate for Essay 1, the monitoring channel in Essay 2, and the moderator quality in Essay 3.
2. **Add the price-learning module to Essay 1.** Test temporal reallocation, information destruction, and depth/composition before adding broad outcome families.
3. **Run the broker-event count audit.** A two-week count determines whether Essay 3 can be identified with adequate power.
4. **Prototype Form 4 outcomes.** Measure the incidence and noise of opportunistic trades before committing to a full historical parse.

Why this order

The first two pilots answer whether the core friction is measurable and whether the independent-treatment essay is feasible. The Form 4 prototype then determines whether the governance paper has a credible primary outcome.

Decisions for the supervisor meeting

My recommendation

Keep “When and Through Which Channel Do Prices Learn?” inside Essay 1. Develop the governance and intermediary-substitution ideas as the two candidate later essays.

1. Is the proposed thesis-level primitive - effective public information depends on discoverability and processing capacity - strong enough to organize all three essays?
2. For Essay 2, is opportunistic insider trading the right primary governance outcome, conditional on passing the first-stage and pre-trend gates?
3. For Essay 3, should the immediate priority be a broker-closure event-count audit before building additional outcomes?
4. What evidence would the committee require before allowing the phrase “firms actively make themselves harder to find”?

Coherence comes from one economic friction, not from reusing one dataset.